Corrected MCQs

1. **B.** The denominator tends to 0 by negative values. The side of the approach determines the sign.
2. **C.** The square root is positive or zero; $\sqrt{x^2} = |x| = -x$ for $x < 0$. This is not an equation to Two solutions.
3. **D.** Divide by x^2 gives $(3 - 1/x)/(1 + 4/x^2) \to 3$.
4. **A.** For $x \neq 1$, the quotient is $x + 1$. A prohibited value does not make a limit impossible.
5. **C.** The gap tends to be 0 and cancels in $k\pi$, $k \geq 1$, with change of sign. An asymptote may to be crossed.
6. **B.** It's x^2/e^x, which tends to 0 by comparison of growth rates. The exponential dominates every power fixed.
7. **A.** Lay down $t = -x \to +\infty$: the expression is $-t/e^t \to 0$.
8. **D.** Multiply by the conjugate quantity gives 2/ $(\sqrt{1 + 2/x} + 1) \to 1$.
9. **C.** With $g = x$, take $f = 1/x^2$, $1/x$ or 1/ $\sqrt{x}$: 0, 1 and $+\infty$.
10. **B.** The comparison theorem is enough for the limit. 2 x^2 refutes the two limits equality; inequality Don't give monotonicity.

True/false. False: Question 5. False: Question 4. False: $x/x \to 1$, $x/x^2 \to 0$, $x^2/x \to +\infty$.

Corrected applications and support

Application 1. In -1^{-} : $+\infty$; in -1^{+} : $-\infty$; both infinites : 3. Asymptotes $x = -1$ and $y = 3$.

Application 2. Limitations: 2; $\frac{4x}{\sqrt{x^2+4x}+x} = \frac{4}{\sqrt{1+4/x}+1} \to 2$; $\frac{1}{\sqrt{1+1/x}+1} \to 1/2$. The expressions are: defined for x big enough.

Implementation 3. For example $x^2/x \to 0$, $x/x \to 1$, $x/x^2 = 1/x \to +\infty$ in 0^{+}. The functions are considered outside 0.

Implementation 4. The first tends to 0 by linearity and comparison of growth rates. For the second, $t = -x \to +\infty$ gives $xe^{x} = -te^{-t} \to 0$. The symbol ∞ is not a real in which one evaluates a function.

Application 5. $q(0) = 1$. $q(x) - 1$ is negative for $x < 0$, positive for $x > 0$. The limits relate to: the infinite ends of the domain, so do not contradict this intersection.

Implementation 6. The limit to infinity is a, therefore $a = 2$; $h(0) = -b = 3$, so $b = -3$. Thus $h(x) = 2 - 1/(x-1)$: limits $+\infty$ in 1^{-}, $-\infty$ in 1^{+} ; curve above 2 left of 1 and in Under the right.

Reading then calculation

Before any table, make the quotient sign anticipate. A numerical result contrary to the sign Expected must trigger a check. p and q on the same window to contest The image of an "inconstructible barrier".

Conduct and correction

Benchmarks for the teacher

In S21, connect each limit to a graphical reading. In S23, require the signs of the denominator left and right. In S24, have the transformation chosen before calculating: factorization, conjugation, local development not used. In S26, separate demonstration from growths Oblique asymptotes are not required here.

Corrected activity

1. Like 3 $x-2=3(x-1)+1$, for $x\neq 1$, $f(x)=3+\frac{1}{x-1}$.

2. In $\pm\infty$, $\frac{1}{x-1}\to 0$: the limits are 3 in 1 $^-$, $x-1\to 0^-$, therefore $f(x)\to-\infty$; in 1 $^+$, $x-1\to 0^+$, so $f(x)\to+\infty$.

3. The sign of $f(x)-3=\frac{1}{x-1}$ is negative for $x<1$, positive for $x>1$. The curve is therefore below and above asymptote, respectively $y=3$.

4. For $x>1$, $|f(x)-3|<0.01$ is equivalent to $\frac{1}{x-1}<0.01$, then $x>101$. We can choose $A=101$.

5. No: $f(x)-3=\frac{1}{x-1}$ The curve may appear to be confused with the asymptote at the precision of the screen, but it does not meet it.

E1. For $x\neq 0$, the quotient $\frac{5-\frac{1}{x^2}}{2+\frac{7}{x^3}}$. In $+\infty$, the inverse powers tend to 0 and the denominator to 2 $\neq 0$. The limit is $\frac{5}{2}$.

E2. In 2, $(x-2)^2\to 0$ by strictly positive values on both sides; its inverse therefore tends towards $+\infty$. In $\pm\infty$, the square tends towards $+\infty$ and its inverse towards 0.

E3. For $x\neq 1$, $x^2-1=(x-1)(x+1)$So the quotient is equal to $x+1$. Its limit in 1 is therefore 2. This simplification calculates the limit; it does not render the initial function defined in 1.

E4. Poisons $t=2x\to+\infty$. Then $x^3e^{-2x}=\frac{t^3}{8e^t}\to 0$, by comparative growth of the exponential and the power.

E5. The conjugate gives $\sqrt{x^2+1}+x=\frac{1}{\sqrt{x^2+1}-x}$. Where $x\to-\infty$, both terms $\sqrt{x^2+1}$ and $-x$ tend to $+\infty$; the denominator therefore tends to $+\infty$. The limit is 0, by positive values.

Corrected from DM3

1. Like $x^2+1>0$ on $\mathbb{R}$, the domain is $\mathbb{R}$. By dividing by x^2 for $x\neq 0$, $f_a(x)=\frac{a+\frac{1}{x^2}}{1+\frac{1}{x^2}}\to a$ in $\pm\infty$.

2. $f_a(x)-a=\frac{ax^2+1-a(x^2+1)}{x^2+1}=\frac{1-a}{x^2+1}$. For $a<1$, the curve is above $y=a$; for $a>1$, it is below; for $a=1$, it is confused with this line.

3. By quotient, $f_a'(x)=\frac{[2ax(x^2+1)-2x(ax^2+1)]}{(x^2+1)^2}=\frac{2x(a-1}{(x^2+1)^2}$. If $a>1$, the sign is that of x : decrease on $]-\infty,0]$, growth on $[0,+\infty[$; minimum $f_a(0)=1$ and no maximum (the bound a is not (a) If $a<1$, the variations are reversed: maximum 1 and no minimum. $a=1$, the function is constant, value 1.

4. For $a=3$ and $x\geq 0$, inequality becomes $\frac{2}{x^2+1}\leq 0{,}02$. The denominator is positive, so it equals to $x^2+1\geq 100$, i.e. $x\geq\overline{99}$.

Corrected work in the manual or an anal

Exercise 26. When $x \to +\infty$, functions $f(x) = x$ and $g(x) = x$ both tend towards $+\infty$, But their difference is no good. $f(x) = 2x$, $g(x) = x$, the difference tends to $+\infty$; by exchanging f and g, it tends towards $-\infty$. These examples establish the indeterminate nature of the difference.

Fiscal year 29. The denominator cancels in $\frac{3}{2}$, therefore $D_f = \mathbb{R} \setminus \{\frac{3}{2}\}$. We're writing

$$f(x) = \frac{5-2x}{2x-3} = -1 + \frac{2}{2x-3}.$$

So $f(x) \to -1$ to the two infinites. $\frac{3}{2}$, the numerator of the second term is positive and its denominator tend towards 0: the limit on the left is $-\infty$, the right is $+\infty$. Asymptotes have as equations $y = -1$ and $x = \frac{3}{2}$.